Wafer edge focus-leveling control in lithography scanners

The central challenge of edge die exposure is not merely missing sensor data — it is the interaction between rapid edge roll-off curvature (quantified by ZDD, the second height derivative) and the scanner's finite leveling bandwidth that fundamentally limits focus accuracy at the wafer periphery. All three major scanner OEMs — ASML, Canon, and Nikon — address this through distinct but converging strategies: pre-exposure height mapping, dynamic sensor selection, and increasingly, external wafer geometry feed-forward. This report synthesizes patent claims, SPIE publications, and simulation frameworks into an engineering-grade reference for building edge-aware focus-leveling systems.

The practical significance is substantial. Samsung demonstrated a **50% reduction in edge defocus** and **1.5 percentage-point yield gain** using ASML's holistic edge leveling optimization on NXT:1960 scanners SPIE Digital Library ResearchGate at 2x-nm memory nodes (Hasan et al., SPIE 9778, 2016). Micron showed that simulation-based optimization of edge clearance settings, using only existing height map data without additional wafer exposures, produces verifiable improvements across DRAM and NAND products SPIE Digital Library Semantic Scholar (Lamonds et al., JM3 18(4), 2019). Semantic Scholar +2 These results confirm that significant edge yield is recoverable through algorithmic improvements alone.

How each OEM architecturally handles edge measurement gaps

ASML: layered defense from hardware through algorithms

ASML's TWINSCAN dual-stage architecture separates measurement from exposure. mit +2 On the measurement stage, a level sensor (LS) with **7–9 optical spots arranged perpendicular to the scan direction** scans the wafer field-by-field, building a complete height map Google Patents before the wafer table swaps to the exposure station. Google Patents The system's edge handling operates through multiple layers:

Global Level Contour Scan (patented in US7,019,815B2, priority 1999). Before detailed height mapping, a single measurement spot traces a path around the wafer perimeter interior. This coarse scan serves two critical functions: predicting the wafer height at scan "in-points" (where the level sensor enters each field column) using **interpolation or extrapolation from nearby data**, and predicting Ry tilt at these in-points to ensure all spots start within their linear range. The patent explicitly states that "only a few data points are required… and they need not be particularly close to the in-points." Google Patents

Height Map Extrapolation Using Chuck Average Maps (patented in US7,649,635B2, priority 2003). When level sensor spots fall off the wafer edge or into mouse-bite regions, this method fills missing data using a pre-characterized average height map of the chuck/support structure. The key insight: the chuck fingerprint needs to be measured only once per chuck, not per wafer batch. Google Patents Combined with available wafer data, this enables "more accurate extrapolation" into unmapped edge regions Google Patents than pure mathematical extension.

AGILE (Air Gauge Improved Process Leveling). A pneumatic focus sensor that measures wafer height independently of the optical level sensor. Because it uses air pressure rather than NIR light, AGILE is immune to process-stack transparency effects that corrupt optical LS readings on back-end-of-line layers. (ResearchGate) (ResearchGate) ASML demonstrated **up to 50% reduction in intrafield focus range** on specific layers. (ResearchGate) (ResearchGate) The correction is applied as an offset map calibrating the optical LS. Limitation: throughput penalty from slower air-gauge measurement, and incompatible with EUV vacuum environments.

UVLS-2 (UV Level Sensor, 2nd generation), deployed on NXT:2050i and NXT:1470. By switching from NIR to UV wavelength, the sensor minimizes film-stack penetration effects. ASML specifically states UVLS-2 provides "increased measurement density" enabling "greater leveling accuracy at the edge of the wafer." (ASML) This is a direct hardware response to the edge data sparsity problem.

Focus Edge Clearance (FEC) and Circuit-Dependent FEC (CDFEC) are customer-configurable parameters defining how close to the wafer/field edge the LS data is trusted. CDFEC adds layout awareness — different clearances for memory array versus periphery regions. (ResearchGate) (ADS) These settings determine the boundary between "measured" and "extrapolated" zones.

Eliminating height map gaps (patented in US6,975,407B1). When the exposure field is narrower than the LS spot array, gaps appear in the height map. The solution translates the spot array toward one field edge, scans, then translates to the opposite edge and rescans (Google Patents) — directly relevant for partial-width edge fields.

Nikon: straight-line pre-scan plus dynamic sensor switching

Nikon's approach centers on two innovations that work in tandem for edge die control.

Straight Line Autofocus (SL-AF), part of the Streamlign platform since ~2008, uses a wide-beam sensor spanning the full 300mm wafer diameter. During wafer loading onto the measurement stage, SL-AF captures a **complete wafer height map in a single linear pass**. (nikon) This pre-scan is especially valuable for edge shots: the system has full-wafer topography data including the ERO region before any exposure begins, enabling accurate stage entry posture prediction where real-time AF data would be unreliable.

Dynamic Edge-Shot Sensor Switching (Hagiwara et al., SPIE 4691, 2002) is the seminal algorithm for real-time edge die AF control. During slit-scanning of edge fields, the algorithm continuously identifies which oblique-incidence AF spots are on-wafer versus off-wafer at each scan position, and **dynamically reconfigures which spots contribute to the Z/Rx/Ry calculation**. (SPIE Digital Library) This interacts with scan direction optimization — choosing whether to scan toward or away from the wafer edge determines whether the AF look-ahead region has valid data or not. Scanning toward the wafer center (AF spots leading into valid territory) is generally preferred.

Nikon's stage entry posture control for edge fields follows a clear sequence: (1) SL-AF pre-scan establishes full wafer map; (2) for edge shots, pre-measured data sets initial Z/Rx/Ry before scanning begins; (3) during scan, dynamic sensor switching uses only valid spots; (4) when all AF

spots become invalid (slit extending beyond wafer edge), the system **holds last valid Z/Rx/Ry values or uses extrapolated values** from the pre-measured map.

Performance data confirms effectiveness. The NSR-S622D achieved **11nm PSFM 3σ including edge shots** through a redesigned AF sensor layout. The NSR-S322F (ArF dry) achieved 12nm PSFM 3σ including edge shots.

A notable hardware approach: Nikon documented **chuck shape modification** to reduce edge defocus (SPIE 5755, 2005), physically reshaping the wafer chuck in the outer region to partially compensate for ERO-induced height deviation. Combined with software "disable range" settings for masking invalid AF sensors, edge defocus was measurably reduced at specific wafer clock positions.

Canon: adaptive measurement point lists and dual focus units

Canon's FPA-series scanners take a more structured approach to edge measurement, documented primarily through patents rather than public SPIE papers.

Adaptive Measurement Point Selection (US8,558,986B2, inventor Yoshimura, 2009). The controller generates a primary list of focus measurement points arranged symmetrically about each shot center at predetermined pitch. For edge shots, it generates a **second list excluding points falling outside the valid wafer area**, then switches to a dedicated control mode using only this reduced point set. google This is a clean, deterministic approach — rather than dynamically switching sensors mid-scan like Nikon, Canon pre-computes the valid point set before exposure begins.

Dual Focus Position Measuring Units (US9,639,008B2, 2014). Canon's twin-stage architecture employs two distinct focus measurement systems: an oblique-projection optical system for Z-axis detection, and a second unit using spot displacement on a 2D imaging element. A fourth measuring device at the patterning station enables correction of errors between the measurement and exposure stations. Google Patents google This redundancy provides cross-checking capability particularly valuable at wafer edges where individual sensor readings may be unreliable.

Pattern Offset Compensation separates and subtracts process-induced topography from the focus measurement, ensuring the AF system tracks the actual wafer surface rather than local pattern features. google At edge dies where pattern layout may be incomplete or asymmetric, this prevents the leveling system from introducing artifacts from partial circuit structures.

ZDD is the metric that predicts edge focus accuracy

A critical finding consistently supported across multiple sources: **focus accuracy at the wafer edge depends on ZDD (Z Double Derivative — the second radial derivative of the height profile), not on ERO magnitude itself** ResearchGate SPIE (Tamura et al., SPIE 6924, 2008). Wafers with low ZDD maintain center-equivalent focus accuracy even at the edge. SPIE Digital Library This has direct implications for incoming wafer specifications and for algorithm design.

ZDD captures the surface curvature that the scanner's slit-based leveling physically cannot correct. The scanner's Z/Rx/Ry stage motion corrects piston, tip, and tilt within the slit — effectively a first-order plane. Residual focus error equals the deviation from this best-fit plane. (Google Patents) A surface with constant slope (high ERO but zero ZDD) is perfectly correctable by tilt. A surface with high curvature (high ZDD) generates non-correctable residual proportional to the curvature magnitude times the slit dimension squared.

KLA's edge feature detection algorithm (US8,594,975B2) formalizes this by computing ZDD and ZDDD (third derivative) profiles in polar coordinates Z(r,θ), applying bilateral filtering with adaptive parameters to distinguish real edge features from noise. Detection thresholds use the formula Tr = (user-set ZDD threshold) × (estimated profile noise level). For bumps embedded in the negative-ZDD roll-off region, ZDDD crossing points identify feature boundaries. (google) This processing pipeline produces the quantitative ERO characterization needed for feed-forward compensation.

Feed-forward from wafer geometry tools closes the edge data gap

The most powerful approach for edge focus control is **supplementing incomplete scanner LS data with external wafer geometry measurements** from tools like KLA's PWG5 (Patterned Wafer Geometry) or Zygo/Ametek instruments. The implementation path is well-defined:

Step 1: Measure free-state wafer geometry. PWG5 captures ⩾4 million data points per wafer using dual-sided interferometry (DirectIndustry) with vertical wafer hold (minimizing gravity effects), measuring shape, flatness, nanotopography, and ERO simultaneously (KLA Corporation +2) — including the full edge region that scanner LS cannot access.

Step 2: Extract chuck fingerprint. KLA's method (US10,576,603B2, Vukkadala & Sinha) subtracts free-state wafer geometry from chucked scanner leveling data on reference wafers. The chuck signature needs to be characterized only once per chuck and captures vacuum-chuck-induced deformation. (Google Patents)

Step 3: Predict chucked wafer topography. For production wafers, predicted chucked flatness = free-state geometry + chuck signature. This gives the full-wafer height map including edge regions that the scanner LS cannot measure. (FPO)

Step 4: Compute per-field/per-slit corrections. The wafer-level flatness map is divided into sites matching scanner exposure fields, then each site is further divided into slit-sized rectangles. Each rectangle is independently leveled (least-squares plane fit), producing the slit-by-slit focus correction matching the scanner's actual leveling granularity. (FPO)

Step 5: Feed corrections via APC interface. KLA's 5D Analyzer computes scanner-compatible correction files (per-exposure-field Z/Rx/Ry offsets) delivered to the scanner's Advanced Process Control interface. (DirectIndustry +2)

Step 6: Close the loop. Post-lithography focus and CDU measurements verify feed-forward accuracy. If residuals remain systematic, a feedback term adjusts subsequent corrections.

Google Patents

For edge dies specifically, this feed-forward pipeline is transformative because the external geometry tool has **complete edge coverage** while the scanner LS does not. The feed-forward data fills exactly the gap that causes edge defocus. KLA demonstrated that PWG-measured wafer flatness metrics predict lithographic defocus with high fidelity ResearchGate +2 (Wong et al., SPIE 2016; Lee et al., SPIE 9424E, 2015).

An advanced variant uses **machine-learning prediction models** (US10,545,412B2, KLA) trained on wafer geometry deltas between process steps. Input features include flatness, thickness, shape, first/higher-order derivatives, shape residuals after 2nd-order removal, chip layout, and film-stack properties. Neural networks, random forests, and boosted regression trees were evaluated, with the best model deployed for wafer-specific feed-forward corrections. Google Patents Justia Patents

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Five extrapolation strategies for edge regions with missing data

When external feed-forward is not available, the focus-leveling system must extrapolate from incomplete scanner LS data into unmeasured edge regions. Each strategy has distinct trade-offs:

Ignore missing spots (reduced-aperture fitting). Use only valid LS spots for least-squares plane fitting. The effective measurement baseline shrinks, degrading Z/Rx/Ry estimation accuracy — particularly Rx and Ry, which require spatial leverage across the slit width. Simple but loses accuracy precisely where it matters most.

Constant extrapolation. Hold the last valid measurement value into the unmeasured region. Computationally trivial and stable, but systematically underestimates ERO magnitude because it cannot track the accelerating roll-off slope. Produces a step discontinuity in the height derivative at the extrapolation boundary.

Linear extrapolation. Extend the local slope from the last few valid measurements. Captures first-order roll-off trends but can overshoot or undershoot if the roll-off curvature is significant (which it typically is — that's the ZDD problem). Can amplify errors if the slope estimate is noisy.

Model-based extrapolation using external data. Incorporate wafer geometry feed-forward (PWG, WaferSight) or pre-characterized ERO profiles to predict edge topography. Most accurate approach but requires additional infrastructure. ASML's chuck-average-map extrapolation (US7,649,635B2) is a lighter-weight variant using only scanner-internal data. Google Patents

B-spline fitting with controlled boundary extension. B-splines offer **local support** — each basis function affects only nearby regions — preventing the global oscillation (Runge phenomenon) that plagues high-order polynomial extrapolation. scikit-learn Bi-quartic B-splines (Wu et al., Acta Optica Sinica, 2024) are demonstrated for wafer map reconstruction. Researching For edge extrapolation, boundary conditions can enforce slope continuity at the last valid measurement while assuming monotonic roll-off continuation. This is the recommended mathematical framework when external geometry data is unavailable.

Zernike polynomials, while standard for circular-aperture wavefront fitting, are **poorly suited for edge extrapolation** due to Gibbs phenomenon at sharp roll-off boundaries, inability to represent non-circular edge-die domains, and inherent instability when extrapolating beyond the fitted region.

Simulation methodology for validating edge focus strategies

The Lamonds et al. framework (JM3 18(4), 043505, 2019) provides a validated simulation approach directly applicable to edge focus algorithm development: SPIE Digital Library +2

Input data: Full-wafer height maps from the scanner's measurement stage (or from external geometry tools). These contain the actual wafer topography including ERO. Researching

Slit-scan simulation algorithm. For each scan position y along a die field, the focus error calculation proceeds as:

1. Extract height values h(x, y') within the slit window: x ∈ [-W_slit/2, +W_slit/2], y' ∈ [y - L_slit/2, y + L_slit/2]
2. Fit a least-squares plane: $z = a_0 + a_1x + a_2y'$, where a_0 is the Z setpoint, a_1 corresponds to Ry, and a_2 to Rx
3. Compute point-by-point focus error: FE(x, y') = h(x, y') - fitted plane
4. **MAFE(y)** = mean of FE over the slit area (systematic defocus at this scan position) Researching
5. **MSDFE(y)** = standard deviation of FE over the slit area (non-correctable residual within the slit) Researching

For edge dies, modify step 1 to reflect which LS spots have valid data under each extrapolation strategy being tested. The simulation computes MAFE and MSDFE for each strategy, enabling direct comparison.

Wafer map reconstruction using Wu et al.'s bi-quartic B-spline parametric surface approach provides smooth, motor-controllable Z/Rx/Ry trajectories. Researching The surface $S(u,v) = \Sigma_i \Sigma_j P_{ij} \times N_{i,4}(u) \times N_{j,4}(v)$, where control points are determined by solving interpolation equations with boundary continuity conditions between adjacent die maps. Their reported results show MAFE ranging from **-35.8 to 13.3 nm** and MSDFE from **5.9 to 41.4 nm** — the authors explicitly note that edge die reconstruction remains an open problem. Researching

CDU impact estimation from simulated defocus uses the quadratic CD-defocus relationship: $\Delta CD \propto (\text{defocus})^2$ for small values. ResearchGate More rigorously, PROLITH-calibrated Bossung curves map each defocus value to a CD shift for the specific feature and process. Google Patents Lithoguru The Normalized Image Log-Slope (NILS) at the mask edge quantifies aerial image quality degradation with defocus. Lithoguru Process window analysis plots CD tolerance contours

in focus-exposure space (Lithoguru) — edge die defocus directly reduces the usable process window.

Validation approach: Lamonds et al. demonstrated that simulated die-fine focus errors correlate with measured electrical yield results, confirming the simulation's predictive power without requiring experimental wafer exposure for each parameter combination. (Semantic Scholar) (SPIE Digital Library)

Consolidated patent reference for edge focus-leveling control

The most actionable patents for system design, organized by function:

- **Pre-scan and entry posture**: ASML US7,019,815B2 (global contour scan with edge extrapolation); Nikon's SL-AF architecture
- **Height map gap filling**: ASML US6,975,407B1 (spot array translation for partial fields); (Google Patents) ASML US7,649,635B2 (extrapolation using chuck average maps)
- **Edge sensor adaptation**: Canon US8,558,986B2 (adaptive measurement point lists); Nikon dynamic sensor switching (SPIE 4691, 2002)
- **Process-robust measurement**: ASML AGILE (air-gauge); (ResearchGate) ASML UVLS-2 (UV level sensor)
- **Feed-forward from geometry tools**: KLA US10,576,603B2 (PWG + chuck extraction); KLA US10,545,412B2 (ML-based prediction) (Justia Patents)
- **ERO characterization**: KLA US8,594,975B2 (ZDD/ZDDD with bilateral filtering)
- **Edge optimization methodology**: Hasan et al. (SPIE 9778, 97783G, 2016); (SPIE Digital Library) Lamonds et al. (JM3 18(4), 043505, 2019)

Conclusion

The edge focus-leveling problem is solvable through a combination of strategies that progressively reduce reliance on incomplete real-time sensor data. **Priority one** for a new focus-leveling system should be implementing robust pre-scan mapping (following Nikon's SL-AF concept or ASML's global contour scan approach) to establish full-wafer topography before exposure, enabling model-based stage entry posture for edge fields. **Priority two** is B-spline-based height map reconstruction with controlled edge extrapolation, avoiding Zernike or high-order polynomial fitting. **Priority three** is integrating external wafer geometry feed-forward (the KLA PWG pipeline), which uniquely provides complete edge coverage. For simulation-based development, the Lamonds/Wu framework — slit-scan simulation from height maps with MAFE/MSDFE metrics — provides a validated, experiment-free optimization loop. (SPIE Digital Library +2) The single most impactful specification to control is incoming wafer ZDD rather than ERO magnitude, as ZDD directly predicts non-correctable edge defocus. (ResearchGate +2)